Structured Programming Lab

Bachelor Program Business Informatics

Department of Computer Science

HelloWorld

The first program

Introduction

- the first program we implement is the HelloWord program.
- this very simple program prints “Hello World!”
- in the code example below we see the basic structure of every Java Programm:
  - a class `HelloWorld` that contains the `main()` function
  - when starting a programm the statements in the `main()` function are executed.

HelloWorld - Code

```
1  package HelloWorld;
2
3  /*
4      my first program
5  */
6
7  public class HelloWorld {
8
9      public static void main(String[] args) {
10         System.out.println("Hello World!"); //a print statement
11     }
12
13 }
```

HelloWorld - Output

```
Hello World!
```

HelloWorld - Flowchart

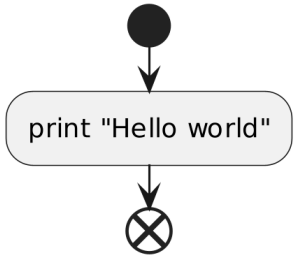


Figure 1: HelloWorld Flow Chart. Note the shapes of the start, end, and output nodes.

HelloWorld - Annotated Flowchart

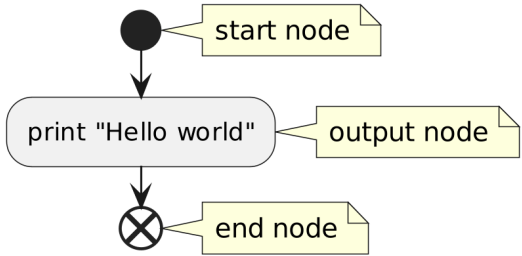


Figure 2: HelloWorld Annotated Flow Chart.

HelloWorld - Walkthrough

Package Declaration

```
1 package HelloWorld;
```

- Java programs are organized in **packages**
- a package is similar to a directory
- in our example, the **class** `HelloWorld` is in the **package** `HelloWorld`
- generally, package and class names are not the same

Comment

```
3  /*
4      my first program
5  */
```

- a **comment** used for documentation for the human reader
- comments start with `/*` and end with `*/`
- everything in between is the comment
- comments are not **compiled** or **executed**, they are meant for documentation

Class Declaration

```
7  public class HelloWorld
```

- name of the class/program: “HelloWorld”

Content of Class

```
7                               {
8       public static void main(String[] args) {
9           System.out.println("Hello World!"); //a print statement
10      }
11
12  }
```

- this **class** contains one single **function**, the `main` function

Main Function

```
9     public static void main(String[] args) {
10        System.out.println("Hello World!"); //a print statement
11    }
```

- **execution** of the program starts in `main` (in line 10)

Content of the `main` Function

```
9                                                   {
10          System.out.println("Hello World!"); //a print statement
11      }
```

- the `main` function has one **statement**
- in `main` we **call** the `println` **function**
- we **pass** the text to the `println` function as a **string** in parenthesis
- statement ends with a semi-colon ;

Line end comment

```
9                                                   {
10                                                  //a print statement
11      }
```

- starts with two forward slashes //
- the rest of the line is ignored
- documentation for the human reader, is not compiled nor executed

String

```
9                                                   {
10                              "Hello World!"
11      }
```

- the **string** (of characters) we want to print
- strings start and end with **double quotes** "
- the string is **delimited** by double quotes